

Richard Teague

rteague@mit.edu

pfl.mit.edu

Kerr-McGee Development Assistant Professor
Massachusetts Institute of Technology

(+1) 617-495-7259
Cambridge, Massachusetts

EDUCATION

- | | |
|-----------|---|
| 2013–2017 | Ph.D. in Astronomy (Magna Cum Laude) — Max-Planck-Institute for Astronomy
Heidelberg, Germany |
| 2008–2013 | MPhys Astrophysics (First Class Honours) — University of Edinburgh
Edinburgh, United Kingdom |

ACADEMIC APPOINTMENTS

- | | |
|---------------|---|
| 07/22–present | Kerr-McGee Development Assistant Professor
Massachusetts Institute of Technology · Dept. of Earth, Atmospheric and Planetary Sciences |
| 05/22–04/25 | Research Associate
Smithsonian Astrophysical Observatory |
| 09/19–04/22 | Submillimeter Array Fellow
Center for Astrophysics · Harvard & Smithsonian |
| 05/17–07/19 | Postdoctoral Researcher
University of Michigan |
| 01/17–04/17 | Postdoctoral Researcher
Max-Planck-Institute for Astronomy |

AWARDS & HONORS

- | | |
|------|--|
| 2026 | AAS Fred Kavli Plenary Speaker — for pioneering work revealing the 3D kinematic velocity fields within planet-forming disks |
| 2024 | EAPS Community Builder Award — recognition of an EAPS community member for exceptional service to EAPS |

09/22	pH Lectureship — recognizes a CfA scientist who shows exceptional promise early in their career
03/20	Harvard Data Science Initiative Research Fund — Regularized Maximum Likelihood Imaging: A New Method for Detecting Planets
11/16	Ernst Patzer Award — awarded for the best refereed publication by a young scientist
2010, 2011	Pre-Honours Certificate of Merit — top 5% performance in pre-honours exams, University of Edinburgh

OBSERVATIONAL TIME & TELESCOPE PROPOSALS (AS PI)

Over 392 hours (600+ as co-I) awarded on ALMA as PI, including the exoALMA Large Program; 20 hours (165 as co-I) on IRAM; 46 hours (30 as co-I) on the SMA; 26 hours (18 as co-I) on JWST as co-PI; 6 hours on VLT/GRAVITY; 3 nights (9 as co-I) on Magellan. Selected PI proposals below.

2021	exoALMA Large Program ALMA, 183 hours, 2021.1.01123.L, A ranked · PI (co-PIs: Benisty, Facchini, Fukagawa & Pinte)
2025	Detecting and Characterizing Magnetic Fields in Two Disks through Zeeman Splitting ALMA, 47 hours, 2025.1.00304.S, B ranked · PI
2025	Calibrating Disk Mass Measurement Techniques ALMA, 14 hours, 2025.1.00940.S, A ranked · PI
2022	The Most Sensitive Search for Magnetic Fields in a Solar Nebula Analogue ALMA, 18 hours, 2022.1.00840.S, A ranked · PI
2021	Is the Magneto-Rotational Instability Driving Protoplanetary Disk Evolution? SMA, 30 hours, 2020A-S033, A ranked · PI
2019	Mapping the 3D Kinematic Structure of Planet Formation ALMA, 33.2 hours, 2019.1.00419.S, B ranked · PI
2018	An Unambiguous Detection of a Magnetic Field in a Protoplanetary Disk ALMA, 6.7 hours, 2018.1.00980.S, A ranked · PI

INVITED TALKS & COLLOQUIA (SELECTED)

60+ invited talks and colloquia since 2015; a selection of recent talks is listed below

03/26	“From Pebbles to Planets: New Frontiers in Planet Formation” — Caltech GPS Division Seminar
02/26	“Searching for Turbulence: Methods, Limitations and Potential” — Flatiron Workshop on Hydrodynamic and Dusty Turbulence in Protoplanetary Disks
01/26	“The Search for Magnetic Fields in Protoplanetary Disks” — MPIA & ESO Star and Planet Formation Seminars
05/25	“An Overview of the exoALMA Program” — COST Action: PLANETS Seminar
04/25–03/25	“How To Build a Solar System” — Johns Hopkins & STScI Joint Colloquium; UC Berkeley Astronomy Colloquium
06/24	“An Interferometric View of the Planet Formation Environment: Past, Present and Future” — GRAVITY+: Impact on Star and Planet Formation
03/24–02/24	“From Pebbles to Planets: What Gas Can Tell Us” — McMaster, CITA & Munich Joint Astronomical Colloquia
12/23	“The Dynamical Structure of Planet Forming Disks” — ALMA at 10 Years: Past, Present, and Future
02/23–06/23	“Witnessing the Formation of Giant Planets and their Moons” — Gordon Conference on the Origins of Solar Systems; Boston University; MIT Haystack; Ohio State; Harvard EPS Colloquia
09/22	“Exploring the Youngest Planetary Systems” — Center for Astrophysics Harvard & Smithsonian pH Lecture
2019–2021	“Witnessing the Assembly of Planetary Systems” / “Mapping the Assembly of Planetary Systems in 6 Dimensions” — CfA, Munich, ETH Zurich, Cambridge, McMaster, Caltech & JPL Colloquia

TEACHING

2022–present	12.410 — Observational Techniques for Optical Astronomy Instructor
2025–present	12.423 — Planet Formation Instructor
2024	12.S681 — From Grains to Gas Giants: The Formation of Planetary Systems Instructor
2014	Wavefront Analysis Laboratory Instructor

ADVISING & MENTORING

Ph.D. students (MIT)	Leah Albrow (2024–present) · Isabella Macias (2024–present) · Jensen Pierce Thomas Lawrence (2023–present)
Masters students	Marco Duraku, Imperial College London (2024–present, MIT–Imperial Exchange) · Amius Marshall De’Ath, Imperial College London (2024–2025) · Abigail Colclasure, MIT (2024–2025)
Undergraduates (MIT)	Jackson Holland (2026–present) · Erin Cusson (2024–present) · Anika Nath (2024) · Carol Chen (2022–2023) · Aidan van Duzer (2023) · Anna Orgel (2022–2023)
Academic advisor	Jan Toomlaid (2024–present) · Kaylee Barrera (2023–present) · Anika Nath (2023–2025)
Prior institutions	Co-supervised graduate and undergraduate students at the University of Michigan (Felipe Alcaron, Jenny Calahan, Deryl Long, Case Hazewinkel, Jeanne Kwon, 2018–2020), Harvard University (Alessandra Canta, 2020–2021), Beijing Normal University (Haochuan Yu, 2020–2022) and Ludwig Maximilian University (Julian Penzinger, 2016, 2018)

PROFESSIONAL SERVICE

Refereeing	Referee for AAS journals, <i>A&A</i> , <i>MNRAS</i> and <i>Nature</i> journals · NESSF external reviewer (2018, 2020)
Departmental (MIT/EAPS)	EAPS Council, Junior Faculty Representative (2025–present) · CORE: Community, Outreach and Respect in EAPS (2025–present) · EAPS Department Lecture Series Planetary Science Representative (2025–present) · EAPS Committee on the Education Program (2023–2024) · EAPS Distinguished Postdoctoral Fellowship Committee (2022)
Workshop & conference SOCs	Astrochemistry in the Broadband Era: ngVLA and ALMA WSU (2025) · Spatio-spectral Modeling of ALMA Data Cubes: ALMA-2030 (2024) · exoALMA Start of Science Workshop (2022) · Vertical Shear Instability Meeting (2022) · SMA Interferometry School (2021) · Visualizing the Kinematics of Planet Formation, Flatiron Institute (2019)
Diversity, equity & inclusion	Postdoc and Research Scientist DEI Representative (2018–2019) · Conversations on Equity and Inclusion Co-organizer (2018–2019) · Equi-Tea Organizer (2018–2019)
Seminar organizing	SMA Seminar Organizer (2020–2021) · Stars, Planets and Formation Seminar Organizer, UMich (2018–2019) · MPIA Student Representative & Workshop Organizer (2015–2017) · IMPRS Graduate Student Representative (2013–2017)

REFEREED PUBLICATIONS

179 refereed articles · 24 as first author · reverse chronological · name in bold

2026

20 PAPERS

- 179 **First Detection of HC₅N in a Class II Disk around TW Hya**
Wampler, S. C., R. A. Loomis, A. Diop, et al.
The Astrophysical Journal, 1003, L30
- 178 **The circumbinary disc of HD 34700A: II. Analysis of a strong dust asymmetry**
Fasano, D., M. Benisty, J. Stadler, et al.
Astronomy and Astrophysics, 709, A174
- 177 **An SMA Molecular Inventory of the Edge-on Protoplanetary Disk Gomez's Hamburger**
Cusson, E. M., L. Wölfer, **R. Teague**, et al.
arXiv e-prints, arXiv:2605.26512
- 176 **Probing Dust in the MWC 480 Disk from Millimeter to Centimeter Wavelengths**
Shi, Y., F. Long, E. Macías, et al.
The Astrophysical Journal, 1001, 27
- 175 **exoALMA. XXI. The Morphology and Dynamics of Vertical Flows**
Benisty, M., A. F. Izquierdo, J. Stadler, et al.
The Astrophysical Journal, 1000, L14
- 174 **exoALMA. XXIII. Estimating Disk and Planet Properties from Dust Morphologies with DBNets 2.0**
Ruzza, A., G. Lodato, G. Rosotti, et al.
The Astrophysical Journal, 1000, L16
- 173 **exoALMA. XX. Tomographic Detection of Embedded Planets in Protoplanetary Disks**
Izquierdo, A. F., J. Bae, S. Facchini, et al.
The Astrophysical Journal, 1000, L13
- 172 **exoALMA XXII: A Two-dimensional Atlas of Deviations from Keplerian Disks**
Fukagawa, M., A. F. Izquierdo, J. Stadler, et al.
The Astrophysical Journal, 1000, L15

- 171 **exoALMA. XXIV. Formaldehyde Emission in Protoplanetary Disks of exoALMA Compared with Their Properties and Dynamical State**
Alarcón, F., S. Facchini, L. Trapman, et al.
The Astrophysical Journal, 1000, L32
- 170 **AB Aur, a Rosetta stone for studies of planet formation: IV. C/O estimates from CS and SO interferometric observations**
Rivière-Marichalar, P., A. Fuente, R. le Gal, et al.
Astronomy and Astrophysics, 707, A348
- 169 **Benchmarking pre-main sequence stellar evolutionary tracks using disk-based dynamical stellar masses**
Zallio, L., M. Vioque, S. M. Andrews, et al.
Astronomy and Astrophysics, 708, L1
- 168 **The circumbinary disk of HD34700A: I. CO gas kinematics indicate spirals, infall, and vortex motions**
Stadler, J., M. Benisty, F. Zagaria, et al.
Astronomy and Astrophysics, 707, A160
- 167 **A Protoplanet Candidate in the PDS 66 Disk Indicated by Silicon Sulfide Isotopologues**
Yoshida, T. C., F. Alarcón, J. Bae, et al.
The Astrophysical Journal, 999, L22
- 166 **A 2 au Resolution View by ALMA of the Planet-hosting WISPIT 2 Disk**
Facchini, S., P. Curone, M. Benisty, et al.
The Astrophysical Journal, 998, L16
- 165 **exoALMA. XIX. Confirmation of Non-thermal Line Broadening in the DM Tau Protoplanetary Disk**
Hardiman, C., C. Pinte, D. J. Price, et al.
The Astrophysical Journal, 997, L47
- 164 **Tracing Pebble Drift History in Two Protoplanetary Disks with CO Enhancement**
Armitage, T., J. Williams, K. Zhang, et al.
The Astrophysical Journal, 998, 308
- 163 **A Submillimeter Survey of CS Excitation in Protoplanetary Disks: Evidence of X-Ray-driven Sulfur Chemistry**
Law, C. J., R. Le Gal, K. I. Öberg, et al.
The Astrophysical Journal, 997, 91
- 162 **Astrometric view of companions in the inner dust cavities of protoplanetary discs**
Vioque, M., R. A. Booth, E. Ragusa, et al.
Astronomy and Astrophysics, 705, A238

- 161 **The ^{12}CO gas structures of protoplanetary disks in the Upper Scorpius region**
Zallio, L., G. P. Rosotti, M. Vioque, et al.
Astronomy and Astrophysics, 705, A49
- 160 **Physical and Chemical Characterization of GY 91's Multi-ringed Protostellar Disk with ALMA**
Jiang, S. D., J. Huang, I. Czekala, et al.
arXiv e-prints, arXiv:2601.18884
- 159 **Direct Measurement of Extinction in a Planet-hosting Gap**
Cugno, G., S. Facchini, F. Alarcon, et al.
The Astronomical Journal, 170, 317
- 158 **Satellites and Small Bodies With ALMA: Insights Into Solar System Formation and Evolution**
de Kleer, K., M. E. Brown, M. Cordiner, et al.
AGU Advances, 6, e2025AV001778
- 157 **Spirals and Vertical Motions in the Planet-Forming Disk around HD 100546. A multi-line study of its gas kinematics**
Wölfer, L., A. F. Izquierdo, A. Booth, et al.
arXiv e-prints, arXiv:2512.13814
- 156 **Winding motion of spirals in a gravitationally unstable protoplanetary disk**
Yoshida, T. C., H. Nomura, K. Doi, et al.
Nature Astronomy, 9, 1672
- 155 **JWST-MIRI Observations of the Irradiated Chemistry in the Inner Disk Cavity of GM Aur**
Romero-Mirza, C. E., K. I. Öberg, A. Banzatti, et al.
The Astrophysical Journal, 991, 128
- 154 **The Light Echo of a High-redshift Quasar Mapped with Ly α Tomography**
Eilers, A.-C., M. Yue, J. Matthee, et al.
The Astrophysical Journal, 991, L40
- 153 **Revealing Fine Structure in Protoplanetary Disks with Physics Constrained Neural Fields**
Levis, A., N. Luong, R. Teague, et al.
arXiv e-prints, arXiv:2509.03623
- 152 **A Radially Resolved Magnetic Field Threading the Disk of TW Hya**
Teague, R., B. Lankhaar, S. M. Andrews, et al.
The Astrophysical Journal, 991, L6

- 151 **exoALMA. XVIII. Interpreting Large-scale Kinematic Structures as Moderate Warping**
Winter, A. J., M. Benisty, A. F. Izquierdo, et al.
The Astrophysical Journal, 990, L10
- 150 **Leaky dust trap in the PDS 70 disc revealed by ALMA Band 9 observations**
Sierra, A., M. Benisty, P. Pinilla, et al.
Monthly Notices of the Royal Astronomical Society, 541, 3101
- 149 **SO Emission in the Dynamically Perturbed Protoplanetary Disks around CQ Tau and MWC 758**
Zagaria, F., H. Jiang, G. Cataldi, et al.
The Astrophysical Journal, 989, 30
- 148 **Inner disc and circumplanetary material in the PDS 70 system: Insights from multi-epoch, multi-frequency ALMA observations**
Fasano, D., M. Benisty, P. Curone, et al.
Astronomy and Astrophysics, 699, A373
- 147 **Radial variations in the nitrogen, carbon, and hydrogen fractionation in the PDS 70 planet-hosting disk**
Rampinelli, L., S. Facchini, M. Leemker, et al.
Astronomy and Astrophysics, 698, A115
- 146 **exoALMA. XIV. Gas Surface Densities in the RX J1604.3–2130 A Disk from Pressure-broadened CO Line Wings**
Yoshida, T. C., P. Curone, J. Stadler, et al.
The Astrophysical Journal, 984, L19
- 145 **exoALMA. XV. Interpreting the Height of CO Emission Layer**
Rosotti, G. P., C. Longarini, T. Paneque-Carreño, et al.
The Astrophysical Journal, 984, L20
- 144 **exoALMA. XVI. Predicting Signatures of Large-scale Turbulence in Protoplanetary Disks**
Barraza-Alfaro, M., M. Flock, W. Béthune, et al.
The Astrophysical Journal, 984, L21
- 143 **exoALMA. XVII. Characterizing the Gas Dynamics around Dust Asymmetries**
Wölfer, L., M. Barraza-Alfaro, R. Teague, et al.
The Astrophysical Journal, 984, L22
- 142 **exoALMA. XII. Weighing and Sizing exoALMA Disks with Rotation Curve Modelling**
Longarini, C., G. Lodato, G. Rosotti, et al.
The Astrophysical Journal, 984, L17

- 141 **exoALMA. V. Gaseous Emission Surfaces and Temperature Structures**
Galloway-Sprietsma, M., J. Bae, A. F. Izquierdo, et al.
The Astrophysical Journal, 984, L10
- 140 **exoALMA. VIII. Probabilistic Moment Maps and Data Products Using Nonparametric Linear Models**
Hilder, T., A. R. Casey, D. J. Price, et al.
The Astrophysical Journal, 984, L13
- 139 **exoALMA. I. Science Goals, Project Design, and Data Products**
Teague, R., M. Benisty, S. Facchini, et al.
The Astrophysical Journal, 984, L6
- 138 **exoALMA. II. Data Calibration and Imaging Pipeline**
Loomis, R. A., S. Facchini, M. Benisty, et al.
The Astrophysical Journal, 984, L7
- 137 **exoALMA. III. Line-intensity Modeling and System Property Extraction from Protoplanetary Disks**
Izquierdo, A. F., J. Stadler, M. Galloway-Sprietsma, et al.
The Astrophysical Journal, 984, L8
- 136 **exoALMA. XI. ALMA Observations and Hydrodynamic Models of LkCa 15: Implications for Planetary Mass Companions in the Dust Continuum Cavity**
Gardner, C. H., A. Isella, H. Li, et al.
The Astrophysical Journal, 984, L16
- 135 **exoALMA. XIII. Gas Masses from N_2H^+ and C^{18}O : A Comparison of Measurement Techniques for Protoplanetary Gas Disk Masses**
Trapman, L., C. Longarini, G. P. Rosotti, et al.
The Astrophysical Journal, 984, L18
- 134 **exoALMA. VI. Rotating under Pressure: Rotation Curves, Azimuthal Velocity Substructures, and Gas Pressure Variations**
Stadler, J., M. Benisty, A. J. Winter, et al.
The Astrophysical Journal, 984, L11
- 133 **exoALMA. VII. Benchmarking Hydrodynamics and Radiative Transfer Codes**
Bae, J., M. Flock, A. Izquierdo, et al.
The Astrophysical Journal, 984, L12
- 132 **exoALMA. IX. Regularized Maximum Likelihood Imaging of Non-Keplerian Features**
Zawadzki, B., I. Czekala, M. Galloway-Sprietsma, et al.
The Astrophysical Journal, 984, L14

- 131 **exoALMA. X. Channel Maps Reveal Complex ^{12}CO Abundance Distributions and a Variety of Kinematic Structures with Evidence for Embedded Planets**
Pinte, C., J. D. Ilee, J. Huang, et al.
The Astrophysical Journal, 984, L15
- 130 **exoALMA. IV. Substructures, Asymmetries, and the Faint Outer Disk in Continuum Emission**
Curone, P., S. Facchini, S. M. Andrews, et al.
The Astrophysical Journal, 984, L9
- 129 **Mapping the Merging Zone of Late Infall in the AB Aur Planet-forming System**
Speedie, J., R. Dong, R. Teague, et al.
The Astrophysical Journal, 981, L30
- 128 **Chemistry in the GG Tau A Disk: Constraints from H_2D^+ , N_2H^+ , and DCO^+ High Angular Resolution ALMA Observations**
Kashyap, P., L. Majumdar, A. Dutrey, et al.
The Astrophysical Journal, 976, 258
- 127 **Kinematical signatures: Distinguishing between warps and radial flows**
Zuleta, A., T. Birnstiel, & R. Teague
Astronomy and Astrophysics, 692, A56
- 126 **Observational signatures of circumbinary discs - II. Kinematic signatures in velocity residuals**
Calcino, J., B. J. Norfolk, D. J. Price, et al.
Monthly Notices of the Royal Astronomical Society, 534, 2904
- 125 **Detection of Dimethyl Ether in the Central Region of the MWC 480 Protoplanetary Disk**
Yamato, Y., Y. Aikawa, V. V. Guzmán, et al.
The Astrophysical Journal, 974, 83
- 124 **Gravitational instability in a planet-forming disk**
Speedie, J., R. Dong, C. Hall, et al.
Nature, 633, 58
- 123 **ALMA high-resolution observations unveil planet formation shaping molecular emission in the PDS 70 disk**
Rampinelli, L., S. Facchini, M. Leemker, et al.
Astronomy and Astrophysics, 689, A65
- 122 **Outflow Driven by a Protoplanet Embedded in the TW Hya Disk**
Yoshida, T. C., H. Nomura, C. J. Law, et al.
The Astrophysical Journal, 971, L15

- 121 **On Kinematic Measurements of Self-gravity in Protoplanetary Disks**
Andrews, S. M., **R. Teague**, C. P. Wirth, et al.
The Astrophysical Journal, 970, 153
- 120 **Constraining the stellar masses and origin of the protostellar VLA 1623 system**
Sadavoy, S. I., P. Sheehan, J. J. Tobin, et al.
Astronomy and Astrophysics, 687, A308
- 119 **The First Spatially Resolved Detection of ^{13}CN in a Protoplanetary Disk and Evidence for Complex Carbon Isotope Fractionation**
Yoshida, T. C., H. Nomura, K. Furuya, et al.
The Astrophysical Journal, 966, 63
- 118 **Mapping the Vertical Gas Structure of the Planet-hosting PDS 70 Disk**
Law, C. J., M. Benisty, S. Facchini, et al.
The Astrophysical Journal, 964, 190
- 117 **The Carbon Isotopic Ratio and Planet Formation**
Bergin, E. A., A. Bosman, **R. Teague**, et al.
The Astrophysical Journal, 965, 147
- 116 **High turbulence in the IM Lup protoplanetary disk. Direct observational constraints from CN and C_2H emission**
Paneque-Carreño, T., A. F. Izquierdo, **R. Teague**, et al.
Astronomy and Astrophysics, 684, A174
- 115 **JWST-MIRI Spectroscopy of Warm Molecular Emission and Variability in the AS 209 Disk**
Romero-Mirza, C. E., K. I. Öberg, A. Banzatti, et al.
The Astrophysical Journal, 964, 36
- 114 **Protoplanetary disks in K_s-band total intensity and polarized light**
Ren, B. B., M. Benisty, C. Ginski, et al.
Astronomy and Astrophysics, 680, A114
- 113 **Three-dimensional magnetic field imaging of protoplanetary disks using Zeeman broadening and linear polarization observations**
Lankhaar, B. & **R. Teague**
Astronomy and Astrophysics, 678, A17
- 112 **MagAO-X and HST High-contrast Imaging of the AS209 Disk at H α**
Cugno, G., Y. Zhou, T. Thanathibodee, et al.
The Astronomical Journal, 166, 162

2023

17 PAPERS

- 111 **MAPS: Constraining Serendipitous Time Variability in Protoplanetary Disk Molecular Ion Emission**
Waggoner, A. R., L. I. Cleeves, R. A. Loomis, et al.
The Astrophysical Journal, 956, 103
- 110 **A Magnetically Driven Disk Wind in the Inner Disk of PDS 70**
Campbell-White, J., C. F. Manara, M. Benisty, et al.
The Astrophysical Journal, 956, 25
- 109 **Constraining the gas distribution in the PDS 70 disc as a method to assess the effect of planet-disc interactions**
Portilla-Revelo, B., I. Kamp, S. Facchini, et al.
Astronomy and Astrophysics, 677, A76
- 108 **Implications for Chondrule Formation Regions and Solar Nebula Magnetism from Statistical Reanalysis of Chondrule Paleomagnetism**
Fu, R. R., S. C. Steele, J. B. Simon, et al.
The Planetary Science Journal, 4, 151
- 107 **Observational signatures of circumbinary discs - I. Kinematics**
Calcino, J., D. J. Price, C. Pinte, et al.
Monthly Notices of the Royal Astronomical Society, 523, 5763
- 106 **Tentative co-orbital submillimeter emission within the Lagrangian region L_5 of the protoplanet PDS 70 b**
Balsalobre-Ruza, O., I. de Gregorio-Monsalvo, J. Lillo-Box, et al.
Astronomy and Astrophysics, 675, A172
- 105 **Kinematic Structures in Planet-Forming Disks**
Pinte, C., R. Teague, K. Flaherty, et al.
Protostars and Planets VII, 534, 645
- 104 **Molecules with ALMA at Planet-forming Scales (MAPS): Complex Kinematics in the AS 209 Disk Induced by a Forming Planet and Disk Winds**
Galloway-Sprietsma, M., J. Bae, R. Teague, et al.
The Astrophysical Journal, 950, 147
- 103 **Mapping Protoplanetary Disk Vertical Structure with CO Isotopologue Line Emission**
Law, C. J., R. Teague, K. I. Öberg, et al.
The Astrophysical Journal, 948, 60
- 102 **An infrared transient from a star engulfing a planet**
De, K., M. MacLeod, V. Karambelkar, et al.
Nature, 617, 55

2022

10 PAPERS

- 101 **A kinematically detected planet candidate in a transition disk**
Stadler, J., M. Benisty, A. Izquierdo, et al.
Astronomy and Astrophysics, 670, L1
- 100 **Kinematics and brightness temperatures of transition discs. A survey of gas substructures as seen with ALMA**
Wölfer, L., S. Facchini, N. van der Marel, et al.
Astronomy and Astrophysics, 670, A154
- 99 **Cold Deuterium Fractionation in the Nearest Planet-forming Disk**
Romero-Mirza, C. E., K. I. Öberg, C. J. Law, et al.
The Astrophysical Journal, 943, 35
- 98 **UV-driven chemistry as a signpost of late-stage planet formation**
Calahan, J. K., E. A. Bergin, A. D. Bosman, et al.
Nature Astronomy, 7, 49
- 97 **A Localized Kinematic Structure Detected in Atomic Carbon Emission Spatially Coincident with a Proposed Protoplanet in the HD 163296 Disk**
Alarcón, F., E. A. Bergin, & R. Teague
The Astrophysical Journal, 941, L24
- 96 **A kinematic excess in the annular gap and gas-depleted cavity in the disc around HD 169142**
Garg, H., C. Pinte, I. Hammond, et al.
Monthly Notices of the Royal Astronomical Society, 517, 5942
- 95 **Unveiling the outer dust disc of TW Hya with deep ALMA observations**
Ilee, J. D., C. Walsh, J. Jennings, et al.
Monthly Notices of the Royal Astronomical Society, 515, L23
- 94 **Mapping the Complex Kinematic Substructure in the TW Hya Disk**
Teague, R., J. Bae, S. M. Andrews, et al.
The Astrophysical Journal, 936, 163
- 93 **Molecules with ALMA at Planet-forming Scales (MAPS): A Circumplanetary Disk Candidate in Molecular-line Emission in the AS 209 Disk**
Bae, J., R. Teague, S. M. Andrews, et al.
The Astrophysical Journal, 934, L20
- 92 **CO Line Emission Surfaces and Vertical Structure in Midinclination Protoplanetary Disks**
Law, C. J., S. Crystian, R. Teague, et al.
The Astrophysical Journal, 932, 114

- 91 **Gas and Dust Shadows in the TW Hydrae Disk**
Teague, R., J. Bae, M. Benisty, et al.
The Astrophysical Journal, 930, 144
- 90 **Polarization from Aligned Dust Grains in the β Pic Debris Disk**
Hull, C. L. H., H. Yang, P. C. Cortés, et al.
The Astrophysical Journal, 930, 49
- 89 **Gas Disk Sizes from CO Line Observations: A Test of Angular Momentum Evolution**
Long, F., S. M. Andrews, G. Rosotti, et al.
The Astrophysical Journal, 931, 6
- 88 **Probing inner and outer disk misalignments in transition disks. Constraints from VLTI/GRAVITY and ALMA observations**
Bohn, A. J., M. Benisty, K. Perraut, et al.
Astronomy and Astrophysics, 658, A183
- 87 **Discovery of Molecular-line Polarization in the Disk of TW Hya**
Teague, R., C. L. H. Hull, S. Guilloteau, et al.
The Astrophysical Journal, 922, 139
- 86 **disksurf: Extracting the 3D Structure of Protoplanetary Disks**
Teague, R., C. Law, J. Huang, et al.
The Journal of Open Source Software, 6, 3827
- 85 **The First Detection of CH₂CN in a Protoplanetary Disk**
Canta, A., R. Teague, R. Le Gal, et al.
The Astrophysical Journal, 922, 62
- 84 **Molecules with ALMA at Planet-forming Scales (MAPS). X. Studying Deuteration at High Angular Resolution toward Protoplanetary Disks**
Cataldi, G., Y. Yamato, Y. Aikawa, et al.
The Astrophysical Journal Supplement Series, 257, 10
- 83 **Molecules with ALMA at Planet-forming Scales (MAPS). XI. CN and HCN as Tracers of Photochemistry in Disks**
Bergner, J. B., K. I. Öberg, V. V. Guzmán, et al.
The Astrophysical Journal Supplement Series, 257, 11
- 82 **Molecules with ALMA at Planet-forming Scales (MAPS). XIII. HCO⁺ and Disk Ionization Structure**
Aikawa, Y., G. Cataldi, Y. Yamato, et al.
The Astrophysical Journal Supplement Series, 257, 13

- 81 **Molecules with ALMA at Planet-forming Scales (MAPS). XVI. Characterizing the Impact of the Molecular Wind on the Evolution of the HD 163296 System**
Booth, A. S., B. Tabone, J. D. Ilee, et al.
The Astrophysical Journal Supplement Series, 257, 16
- 80 **Molecules with ALMA at Planet-forming Scales (MAPS). XIX. Spiral Arms, a Tail, and Diffuse Structures Traced by CO around the GM Aur Disk**
Huang, J., E. A. Bergin, K. I. Öberg, et al.
The Astrophysical Journal Supplement Series, 257, 19
- 79 **Molecules with ALMA at Planet-forming Scales. XX. The Massive Disk around GM Aurigae**
Schwarz, K. R., J. K. Calahan, K. Zhang, et al.
The Astrophysical Journal Supplement Series, 257, 20
- 78 **Molecules with ALMA at Planet-forming Scales (MAPS). VI. Distribution of the Small Organics HCN, C₂H, and H₂CO**
Guzmán, V. V., J. B. Bergner, C. J. Law, et al.
The Astrophysical Journal Supplement Series, 257, 6
- 77 **Molecules with ALMA at Planet-forming Scales (MAPS). VIII. CO Gap in AS 209-Gas Depletion or Chemical Processing?**
Alarcón, F., A. D. Bosman, E. A. Bergin, et al.
The Astrophysical Journal Supplement Series, 257, 8
- 76 **Molecules with ALMA at Planet-forming Scales (MAPS). XVII. Determining the 2D Thermal Structure of the HD 163296 Disk**
Calahan, J. K., E. A. Bergin, K. Zhang, et al.
The Astrophysical Journal Supplement Series, 257, 17
- 75 **Molecules with ALMA at Planet-forming Scales (MAPS). III. Characteristics of Radial Chemical Substructures**
Law, C. J., R. A. Loomis, R. Teague, et al.
The Astrophysical Journal Supplement Series, 257, 3
- 74 **Molecules with ALMA at Planet-forming Scales (MAPS). IV. Emission Surfaces and Vertical Distribution of Molecules**
Law, C. J., R. Teague, R. A. Loomis, et al.
The Astrophysical Journal Supplement Series, 257, 4
- 73 **Molecules with ALMA at Planet-forming Scales (MAPS). XVIII. Kinematic Substructures in the Disks of HD 163296 and MWC 480**
Teague, R., J. Bae, Y. Aikawa, et al.
The Astrophysical Journal Supplement Series, 257, 18

- 72 **Molecules with ALMA at Planet-forming Scales (MAPS). XIV. Revealing Disk Substructures in Multiwavelength Continuum Emission**
Sierra, A., L. M. Pérez, K. Zhang, et al.
The Astrophysical Journal Supplement Series, 257, 14
- 71 **Molecules with ALMA at Planet-forming Scales (MAPS). XV. Tracing Protoplanetary Disk Structure within 20 au**
Bosman, A. D., E. A. Bergin, R. A. Loomis, et al.
The Astrophysical Journal Supplement Series, 257, 15
- 70 **Molecules with ALMA at Planet-forming Scales (MAPS). I. Program Overview and Highlights**
Öberg, K. I., V. V. Guzmán, C. Walsh, et al.
The Astrophysical Journal Supplement Series, 257, 1
- 69 **Molecules with ALMA at Planet-forming Scales (MAPS). II. CLEAN Strategies for Synthesizing Images of Molecular Line Emission in Protoplanetary Disks**
Czekala, I., R. A. Loomis, R. Teague, et al.
The Astrophysical Journal Supplement Series, 257, 2
- 68 **Molecules with ALMA at Planet-forming Scales (MAPS). V. CO Gas Distributions**
Zhang, K., A. S. Booth, C. J. Law, et al.
The Astrophysical Journal Supplement Series, 257, 5
- 67 **Molecules with ALMA at Planet-forming Scales (MAPS). VII. Substellar O/H and C/H and Superstellar C/O in Planet-feeding Gas**
Bosman, A. D., F. Alarcón, E. A. Bergin, et al.
The Astrophysical Journal Supplement Series, 257, 7
- 66 **Molecules with ALMA at Planet-forming Scales (MAPS). IX. Distribution and Properties of the Large Organic Molecules HC₃N, CH₃CN, and c-C₃H₂**
Ilee, J. D., C. Walsh, A. S. Booth, et al.
The Astrophysical Journal Supplement Series, 257, 9
- 65 **Molecules with ALMA at Planet-forming Scales (MAPS). XII. Inferring the C/O and S/H Ratios in Protoplanetary Disks with Sulfur Molecules**
Le Gal, R., K. I. Öberg, R. Teague, et al.
The Astrophysical Journal Supplement Series, 257, 12
- 64 **Mapping the 3D Kinematical Structure of the Gas Disk of HD 169142**
Yu, H., R. Teague, J. Bae, et al.
The Astrophysical Journal, 920, L33
- 63 **The Chemical Inventory of the Planet-hosting Disk PDS 70**
Facchini, S., R. Teague, J. Bae, et al.
The Astronomical Journal, 162, 99

- 62 **Limits on Millimeter Continuum Emission from Circumplanetary Material in the DSHARP Disks**
Andrews, S. M., W. Elder, S. Zhang, et al.
The Astrophysical Journal, 916, 51
- 61 **The Architecture of the V892 Tau System: The Binary and Its Circumbinary Disk**
Long, F., S. M. Andrews, J. Vega, et al.
The Astrophysical Journal, 915, 131
- 60 **A Circumplanetary Disk around PDS70c**
Benisty, M., J. Bae, S. Facchini, et al.
The Astrophysical Journal, 916, L2
- 59 **Investigating the Relative Gas and Small Dust Grain Surface Heights in Protoplanetary Disks**
Rich, E. A., R. Teague, J. D. Monnier, et al.
The Astrophysical Journal, 913, 138
- 58 **Vortex-like kinematic signal, spirals, and beam smearing effect in the HD 142527 disk**
Boehler, Y., F. Ménard, C. M. T. Robert, et al.
Astronomy and Astrophysics, 650, A59
- 57 **Observational Signature of Tightly Wound Spirals Driven by Buoyancy Resonances in Protoplanetary Disks**
Bae, J., R. Teague, & Z. Zhu
The Astrophysical Journal, 912, 56
- 56 **A highly non-Keplerian protoplanetary disc. Spiral structure in the gas disc of CQ Tau**
Wölfer, L., S. Facchini, N. T. Kurtovic, et al.
Astronomy and Astrophysics, 648, A19
- 55 **An Atacama Large Millimeter/submillimeter Array Survey of Chemistry in Disks around M4-M5 Stars**
Pegues, J., K. I. Öberg, J. B. Bergner, et al.
The Astrophysical Journal, 911, 150
- 54 **The TW Hya Rosetta Stone Project IV: A Hydrocarbon-rich Disk Atmosphere**
Cleeves, L. I., R. A. Loomis, R. Teague, et al.
The Astrophysical Journal, 911, 29
- 53 **ALMA CN Zeeman Observations of AS 209: Limits on Magnetic Field Strength and Magnetically Driven Accretion Rate**
Harrison, R. E., L. W. Looney, I. W. Stephens, et al.
The Astrophysical Journal, 908, 141

2020

17 PAPERS

- 52 **The TW Hya Rosetta Stone Project. III. Resolving the Gaseous Thermal Profile of the Disk**
Calahan, J. K., E. Bergin, K. Zhang, et al.
The Astrophysical Journal, 908, 8
- 51 **Dynamical Masses and Stellar Evolutionary Model Predictions of M Stars**
Pegues, J., I. Czekala, S. M. Andrews, et al.
The Astrophysical Journal, 908, 42
- 50 **The TW Hya Rosetta Stone Project. I. Radial and Vertical Distributions of DCN and DCO⁺**
Öberg, K. I., L. I. Cleeves, J. B. Bergner, et al.
The Astronomical Journal, 161, 38
- 49 **The TW Hya Rosetta Stone Project. II. Spatially Resolved Emission of Formaldehyde Hints at Low-temperature Gas-phase Formation**
Terwisscha van Scheltinga, J., M. R. Hogerheijde, L. I. Cleeves, et al.
The Astrophysical Journal, 906, 111
- 48 **ALMA chemical survey of disk-outflow sources in Taurus (ALMA-DOT). V. Sample, overview, and demography of disk molecular emission**
Garufi, A., L. Podio, C. Codella, et al.
Astronomy and Astrophysics, 645, A145
- 47 **ALMA chemical survey of disk-outflow sources in Taurus (ALMA-DOT). III. The interplay between gas and dust in the protoplanetary disk of DG Tau**
Podio, L., A. Garufi, C. Codella, et al.
Astronomy and Astrophysics, 644, A119
- 46 **Chemical Evolution in a Protoplanetary Disk within Planet Carved Gaps and Dust Rings**
Alarcón, F., R. Teague, K. Zhang, et al.
The Astrophysical Journal, 905, 68
- 45 **Predicting the Kinematic Evidence of Gravitational Instability**
Hall, C., R. Dong, R. Teague, et al.
The Astrophysical Journal, 904, 148
- 44 **GoFish: Molecular line detections in protoplanetary disks**
Teague, R.
Astrophysics Source Code Library, ascl:2011.016
- 43 **ALMA and VLA Observations of EX Lupi in Its Quiescent State**
White, J. A., Á. Kóspál, A. G. Hughes, et al.
The Astrophysical Journal, 904, 37

- 42 **Visualizing the Kinematics of Planet Formation**
Disk Dynamics Collaboration, P. J. Armitage, J. Bae, et al.
arXiv e-prints, arXiv:2009.04345
- 41 **Low-level Carbon Monoxide Line Polarization in Two Protoplanetary Disks: HD 142527 and IM Lup**
Stephens, I. W., M. Fernández-López, Z.-Y. Li, et al.
The Astrophysical Journal, 901, 71
- 40 **The Excitation Conditions of CN in TW Hya**
Teague, R. & R. Loomis
The Astrophysical Journal, 899, 157
- 39 **Annular substructures in the transition disks around LkCa 15 and J1610**
Facchini, S., M. Benisty, J. Bae, et al.
Astronomy and Astrophysics, 639, A121
- 38 **A three-dimensional view of Gomez's hamburger**
Teague, R., M. R. Jankovic, T. J. Haworth, et al.
Monthly Notices of the Royal Astronomical Society, 495, 451
- 37 **Hints of a Population of Solar System Analog Planets from ALMA**
Long, D. E., K. Zhang, R. Teague, et al.
The Astrophysical Journal, 895, L46
- 36 **The efficiency of dust trapping in ringed protoplanetary discs**
Rosotti, G. P., R. Teague, C. Dullemond, et al.
Monthly Notices of the Royal Astronomical Society, 495, 173
- 35 **ALMA chemical survey of disk-outflow sources in Taurus (ALMA-DOT). I. CO, CS, CN, and H₂CO around DG Tau B**
Garufi, A., L. Podio, C. Codella, et al.
Astronomy and Astrophysics, 636, A65
- 34 **A Multifrequency ALMA Characterization of Substructures in the GM Aur Protoplanetary Disk**
Huang, J., S. M. Andrews, C. P. Dullemond, et al.
The Astrophysical Journal, 891, 48
- 33 **Spiral arms in the protoplanetary disc HD100453 detected with ALMA: evidence for binary-disc interaction and a vertical temperature gradient**
Rosotti, G. P., M. Benisty, A. Juhász, et al.
Monthly Notices of the Royal Astronomical Society, 491, 1335
- 32 **Tracing the physical conditions of planet formation with molecular excitation**
Teague, R.
Laboratory Astrophysics: From Observations to Interpretation, 350, 181

2019

12 PAPERS

- 31 **Accretion disks around young stars: the cradles of planet formation**
Semenov, D. A. & R. D. Teague
Europhysics News, 51, 29
- 30 **Meridional flows in the disk around a young star**
Teague, R., J. Bae, & E. A. Bergin
Nature, 574, 378
- 29 **Spiral Structure in the Gas Disk of TW Hya**
Teague, R., J. Bae, J. Huang, et al.
The Astrophysical Journal, 884, L56
- 28 **An Ideal Testbed for Planet-Disk Interaction: Two Giant Protoplanets in Resonance Shaping the PDS 70 Protoplanetary Disk**
Bae, J., Z. Zhu, C. Baruteau, et al.
The Astrophysical Journal, 884, L41
- 27 **GoFish: Fishing for Line Observations in Protoplanetary Disks**
Teague, R.
The Journal of Open Source Software, 4, 1632
- 26 **ALMA observations of A0620-00: fresh clues on the nature of quiescent black hole X-ray binary jets**
Gallo, E., R. Teague, R. M. Plotkin, et al.
Monthly Notices of the Royal Astronomical Society, 488, 191
- 25 **Detection of Continuum Submillimeter Emission Associated with Candidate Protoplanets**
Isella, A., M. Benisty, R. Teague, et al.
The Astrophysical Journal, 879, L25
- 24 **Statistical Uncertainties in Moment Maps of Line Emission**
Teague, R.
Research Notes of the American Astronomical Society, 3, 74
- 23 **Realizing the Unique Potential of ALMA to Probe the Gas Reservoir of Planet Formation**
Cleeves, I., R. Loomis, R. Teague, et al.
Bulletin of the American Astronomical Society, 51, 81
- 22 **Planet formation — The case for large efforts on the computational side**
Lyra, W., T. Haworth, B. Bitsch, et al.
Bulletin of the American Astronomical Society, 51, 129
- 21 **Highly structured disk around the planet host PDS 70 revealed by high-angular resolution observations with ALMA**
Keppler, M., R. Teague, J. Bae, et al.
Astronomy and Astrophysics, 625, A118

2018

7 PAPERS

- 20 **Line Ratios Reveal N_2H^+ Emission Originates above the Midplane in TW Hydrae**
Schwarz, K. R., **R. Teague**, & E. A. Bergin
The Astrophysical Journal, 876, L13
- 19 **eddy: Extracting Protoplanetary Disk Dynamics with Python**
Teague, R.
The Journal of Open Source Software, 4, 1220
- 18 **Evidence for a Vertical Dependence on the Pressure Structure in AS 209**
Teague, R., J. Bae, T. Birnstiel, et al.
The Astrophysical Journal, 868, 113
- 17 **Temperature, Mass, and Turbulence: A Spatially Resolved Multiband Non-LTE Analysis of CS in TW Hya**
Teague, R., T. Henning, S. Guilloteau, et al.
The Astrophysical Journal, 864, 133
- 16 **Chemistry in disks. XI. Sulfur-bearing species as tracers of protoplanetary disk physics and chemistry: the DM Tau case**
Semenov, D., C. Favre, D. Fedele, et al.
Astronomy and Astrophysics, 617, A28
- 15 **A Robust Method to Measure Centroids of Spectral Lines**
Teague, R. & D. Foreman-Mackey
Research Notes of the American Astronomical Society, 2, 173
- 14 **A Kinematical Detection of Two Embedded Jupiter-mass Planets in HD 163296**
Teague, R., J. Bae, E. A. Bergin, et al.
The Astrophysical Journal, 860, L12
- 13 **Turbulence in the TW Hya Disk**
Flaherty, K. M., A. M. Hughes, **R. Teague**, et al.
The Astrophysical Journal, 856, 117
- 12 **ALMA continuum observations of the protoplanetary disk AS 209. Evidence of multiple gaps opened by a single planet**
Fedele, D., M. Tazzari, R. Booth, et al.
Astronomy and Astrophysics, 610, A24

2017

7 PAPERS

- 11 **Radiation Hydrodynamical Turbulence in Protoplanetary Disks: Numerical Models and Observational Constraints**
Flock, M., R. P. Nelson, N. J. Turner, et al.
The Astrophysical Journal, 850, 131

- 10 **The Flying Saucer: Tomography of the thermal and density gas structure of an edge-on protoplanetary disk**
Dutrey, A., S. Guilloteau, V. Piétu, et al.
Astronomy and Astrophysics, 607, A130
- 9 **Multiplicity and disks within the high-mass core NGC 7538IRS1. Resolving cm line and continuum emission at $0.06'' \times 0.05''$ resolution**
Beuther, H., H. Linz, T. Henning, et al.
Astronomy and Astrophysics, 605, A61
- 8 **On the methanol emission detection in the TW Hya disc: the role of grain surface chemistry and non-LTE excitation**
Parfenov, S. Y., D. A. Semenov, T. Henning, et al.
Monthly Notices of the Royal Astronomical Society, 468, 2024
- 7 **Three Radial Gaps in the Disk of TW Hydrae Imaged with SPHERE**
van Boekel, R., T. Henning, J. Menu, et al.
The Astrophysical Journal, 837, 132
- 6 **A Surface Density Perturbation in the TW Hydrae Disk at 95 au Traced by Molecular Emission**
Teague, R., D. Semenov, U. Gorti, et al.
The Astrophysical Journal, 835, 228
- 5 **Tracing the earliest stages of planet formation through modelling and sub-mm observations**
Teague, R.
Ph.D. Thesis
- 4 **Grand Challenges in Protoplanetary Disc Modelling**
Haworth, T. J., J. D. Ilee, D. H. Forgan, et al.
Publications of the Astronomical Society of Australia, 33, e053
- 3 **Inferring the evolutionary stages of the internal structures of NGC 7538 S and IRS1 from chemistry**
Feng, S., H. Beuther, D. Semenov, et al.
Astronomy and Astrophysics, 593, A46
- 2 **Measuring turbulence in TW Hydrae with ALMA: methods and limitations**
Teague, R., S. Guilloteau, D. Semenov, et al.
Astronomy and Astrophysics, 592, A49

2016

3 PAPERS

2015

1 PAPERS

¹ Chemistry in disks. IX. Observations and modelling of HCO⁺ and DCO⁺ in DM Tauri

Teague, R., D. Semenov, S. Guilloteau, et al.

Astronomy and Astrophysics, 574, A137